Power stations

Electricity is an extremely versatile source of energy, able to be transported long distances and used in countless applications. Vast amounts of electricity are generated by power stations, most burning fossil fuels such as coal.

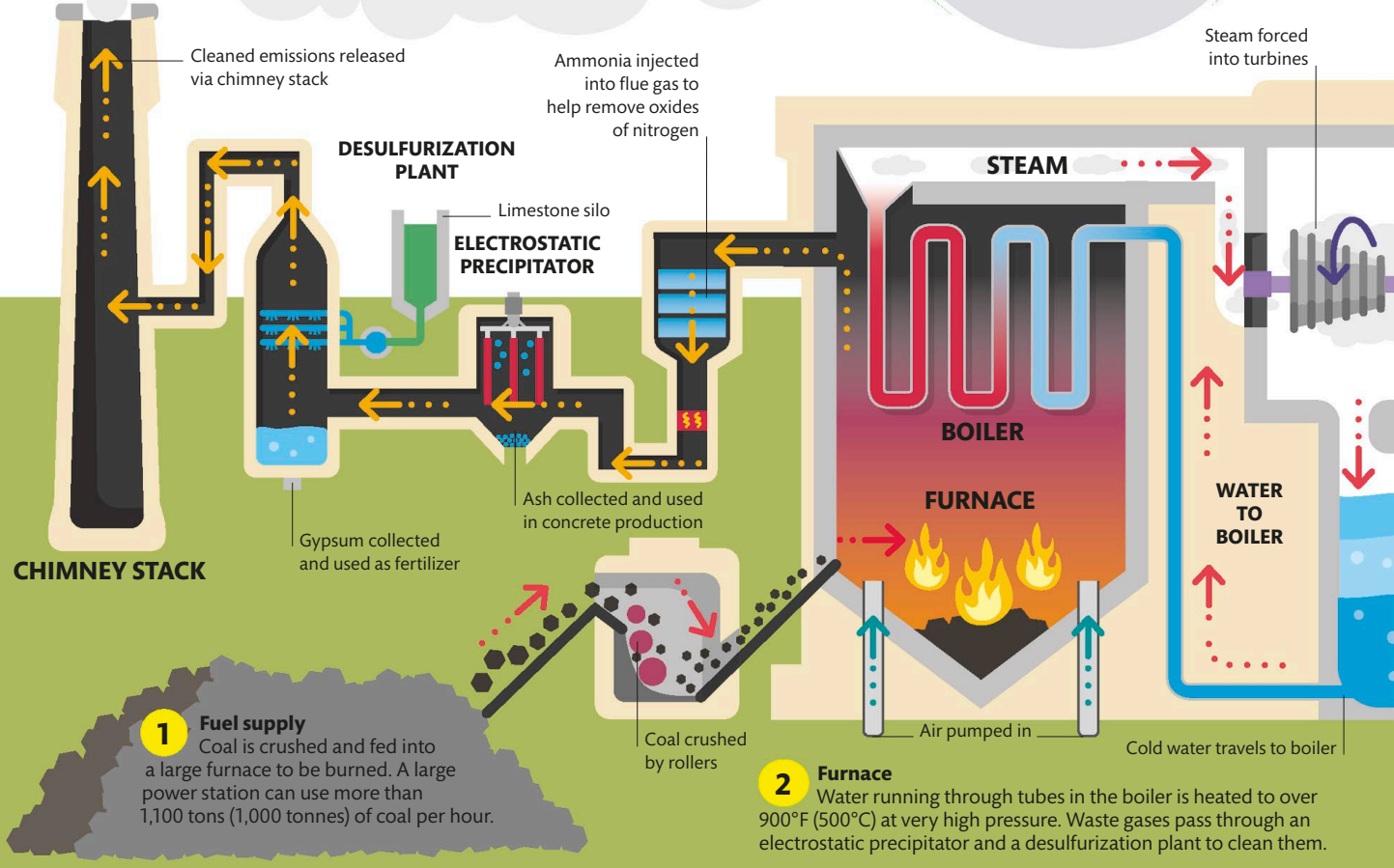
66 PERCENT OF THE WORLD'S ELECTRICITY SUPPLY COMES FROM FOSSIL FUELS

How a power station works

In a conventional coal-fired power station, a furnace heats water to create superheated steam. This drives a turbine that, in turn, powers electricity generators. A large power station can generate 2,000 megawatts of electricity—enough to power more than a million households. The used steam is cooled, condensed back to water, and reused; waste gases are treated and cleaned; and the furnace's ash is often processed into cinder blocks.

HAS OUR RELIANCE ON COAL DECREASED?

On the contrary, coal use has been rocketing in recent decades. Since the 1970s, our annual consumption has increased by more than 200 percent.



1 Fuel supply
Coal is crushed and fed into a large furnace to be burned. A large power station can use more than 1,100 tons (1,000 tonnes) of coal per hour.

2 Furnace
Water running through tubes in the boiler is heated to over 900°F (500°C) at very high pressure. Waste gases pass through an electrostatic precipitator and a desulfurization plant to clean them.